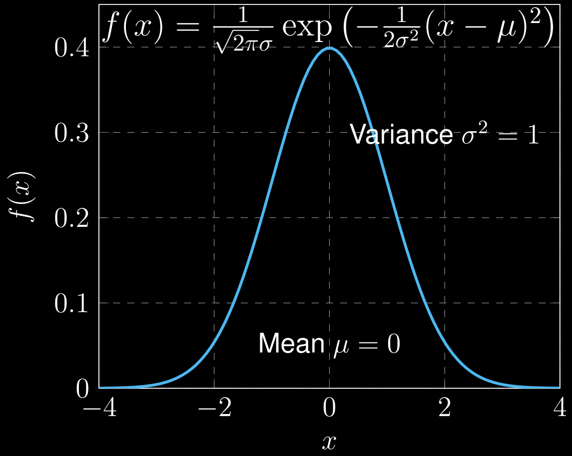
Standard Gaussian Distribution



Multivariate Gaussian Distribution

Probability density function:

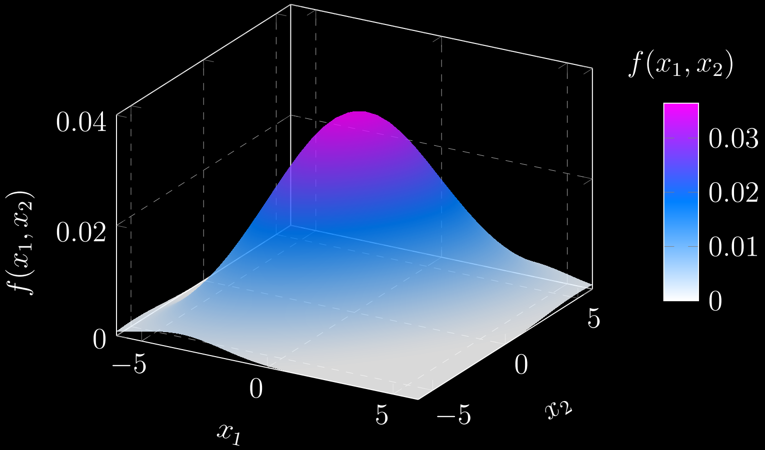
$$f(\boldsymbol{x}) = \frac{1}{(2\pi)^{n/2}\sqrt{\det(\boldsymbol{\Sigma})}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)$$

with mean vector $\boldsymbol{\mu} \in \mathbb{R}^n$ and covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$.

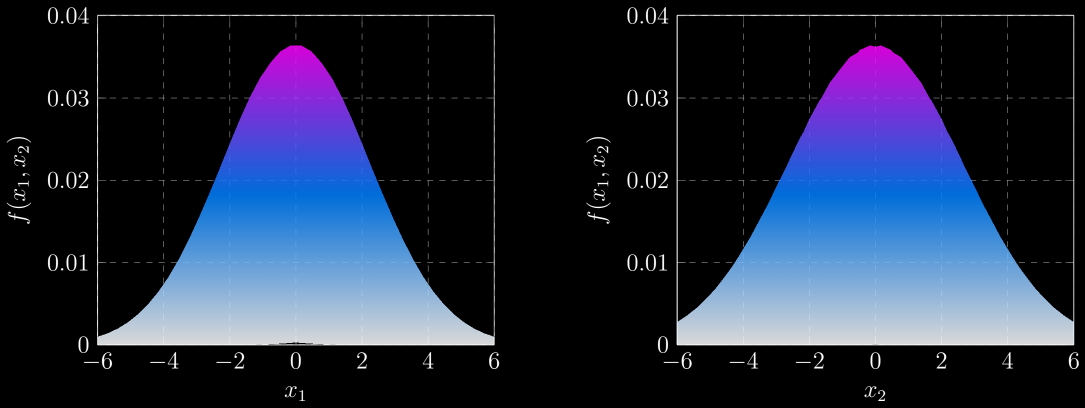
Bivariate Gaussian distribution:

Mean vector $\boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and covariance matrix $\boldsymbol{\Sigma} = \begin{bmatrix} 5 & 4 \\ 4 & 7 \end{bmatrix}$

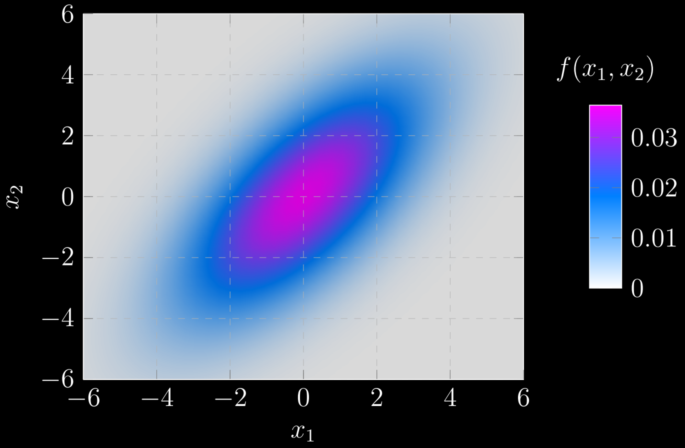
Bivariate Gaussian Distribution



Bivariate Gaussian Distribution



Bivariate Gaussian Distribution



The level set of a Gaussian distribution is the set of points in n-dimensional space with the same probability density.

$$\frac{1}{(2\pi)^{n/2}\sqrt{\det(\boldsymbol{\Sigma})}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right) = c_0 \quad \text{(at level } c_0\text{)}$$

$$\Rightarrow \qquad (\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}) = c \quad \text{(with constant } c\text{)}$$

The level set of multivariate Gaussian distribution is an ellipsoid in the n-dimensional space, why?

$$
\begin{aligned}
&(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}) \\
=&(\boldsymbol{x}-\boldsymbol{\mu})^\top \left(\sum_{i=1}^{n} \lambda_i \boldsymbol{u}_i \boldsymbol{u}_i^\top\right)^{-1}(\boldsymbol{x}-\boldsymbol{\mu})
\end{aligned}
$$

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=&(\boldsymbol{x}-\boldsymbol{\mu})^\top \left(\sum_{i=1}^{n} \frac{1}{\lambda_i} \boldsymbol{u}_i \boldsymbol{u}_i^\top\right)(\boldsymbol{x}-\boldsymbol{\mu}) = c
\end{aligned}
$$

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=&(\boldsymbol{x}-\boldsymbol{\mu})^\top \left(\sum_{i=1}^{n} \frac{1}{\lambda_i} \boldsymbol{u}_i \boldsymbol{u}_i^\top\right)(\boldsymbol{x}-\boldsymbol{\mu}) = c
\end{aligned}
$$

$$
\Rightarrow \quad \frac{(\boldsymbol{u}_1^\top(\boldsymbol{x}-\boldsymbol{\mu}))^2}{\lambda_1} + \cdots + \frac{(\boldsymbol{u}_n^\top(\boldsymbol{x}-\boldsymbol{\mu}))^2}{\lambda_n} = c \qquad \text{(ellipsoid)}
$$

with eigenvalue λ_i and eigenvector $\boldsymbol{u}_i$.

Bivariate Gaussian distribution:

Mean vector $\boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and covariance matrix $\boldsymbol{\Sigma} = \begin{bmatrix} 5 & 4 \\ 4 & 7 \end{bmatrix}$

The ellipse:

$$\frac{(\boldsymbol{u}_1^\top \boldsymbol{x})^2}{\lambda_1} + \frac{(\boldsymbol{u}_2^\top \boldsymbol{x})^2}{\lambda_2} = c$$

with eigenvalues λ_1, λ_2 and eigenvectors $\boldsymbol{u}_1, \boldsymbol{u}_2$.

Covariance Matrix $\mathbf{\Sigma} = \begin{bmatrix} 5 & 4 \\ 4 & 7 \end{bmatrix} \in \mathbb{R}^{2\times 2}$

Eigenvalue decomposition:

$$\mathbf{\Sigma} = \lambda_1 \boldsymbol{u}_1 \boldsymbol{u}_1^\top + \lambda_2 \boldsymbol{u}_2 \boldsymbol{u}_2^\top$$

with eigenvalues:

$$\lambda_1 = 1.877 \quad \lambda_2 = 10.123$$

and eigenvectors:

$$\boldsymbol{u}_1 = \begin{bmatrix} -0.788 \\ 0.615 \end{bmatrix} \quad \boldsymbol{u}_2 = \begin{bmatrix} -0.615 \\ -0.788 \end{bmatrix}$$

Covariance Matrix $\boldsymbol{\Sigma} = \begin{bmatrix} 5 & 4 \\ 4 & 7 \end{bmatrix} \in \mathbb{R}^{2\times 2}$

Eigenvalue decomposition:

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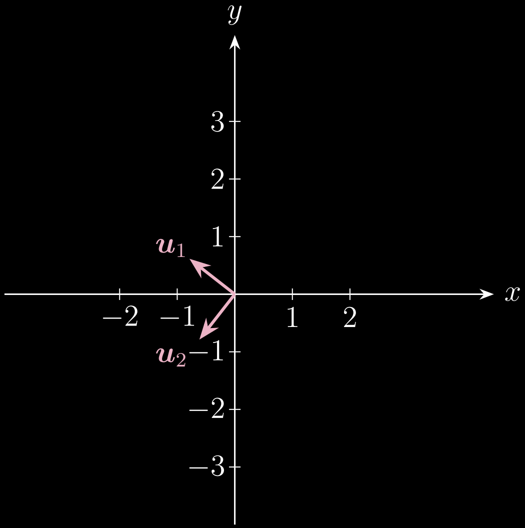
and eigenvectors:

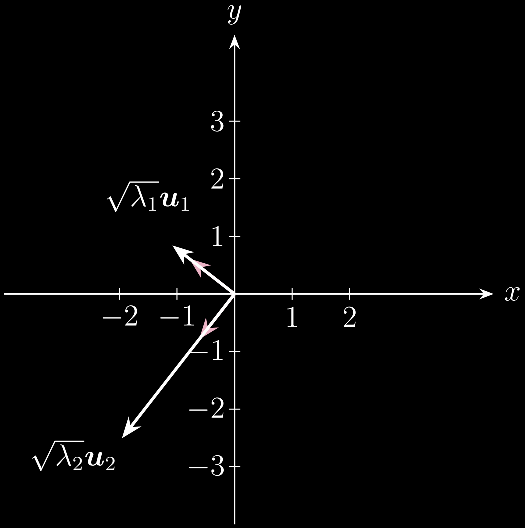
$$\boldsymbol{u}_1 = \begin{bmatrix} -0.788 \\ 0.615 \end{bmatrix} \quad \boldsymbol{u}_2 = \begin{bmatrix} -0.615 \\ -0.788 \end{bmatrix}$$

```python
import numpy as np

S = np.array([[5, 4], [4, 7]])
eig_val, eig_vec = np.linalg.eig(S)

print('lambda_1 = ', eig_val[0])
print('lambda_2 = ', eig_val[1])
print('u_1 = ', eig_vec[:, 0])
print('u_2 = ', eig_vec[:, 1])
```

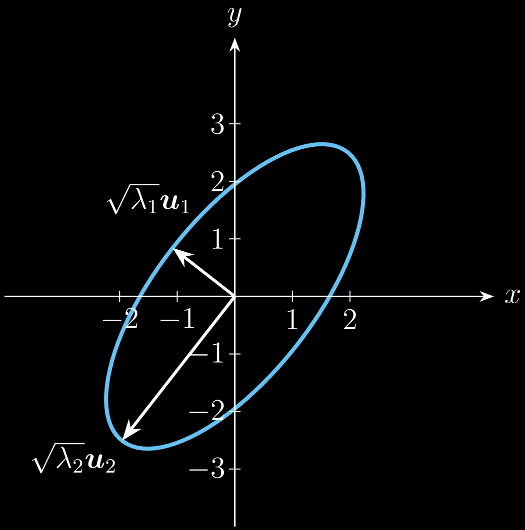
Ellipse $\dfrac{(\boldsymbol{u}_1^\top \boldsymbol{x})^2}{(\sqrt{\lambda_1})^2} + \dfrac{(\boldsymbol{u}_2^\top \boldsymbol{x})^2}{(\sqrt{\lambda_2})^2} = 1$



Ellipse $\dfrac{(\boldsymbol{u}_1^\top \boldsymbol{x})^2}{(\sqrt{\lambda_1})^2} + \dfrac{(\boldsymbol{u}_2^\top \boldsymbol{x})^2}{(\sqrt{\lambda_2})^2} = 1$

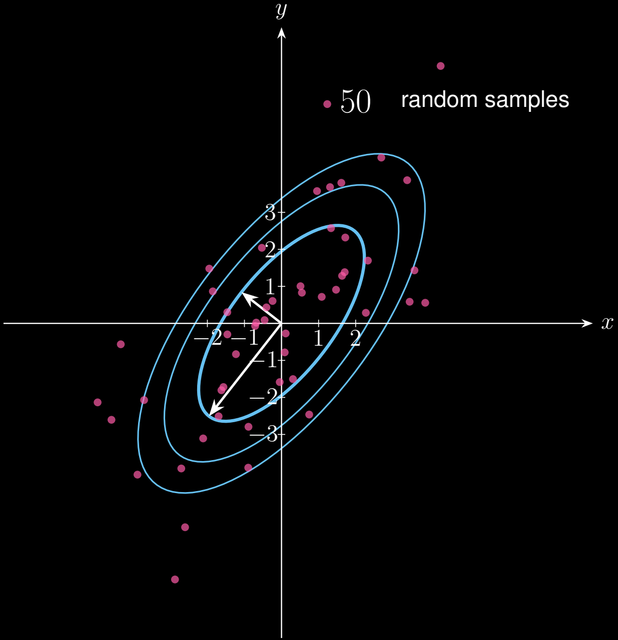


Ellipse $\dfrac{(\boldsymbol{u}_1^\top \boldsymbol{x})^2}{(\sqrt{\lambda_1})^2} + \dfrac{(\boldsymbol{u}_2^\top \boldsymbol{x})^2}{(\sqrt{\lambda_2})^2} = 1$

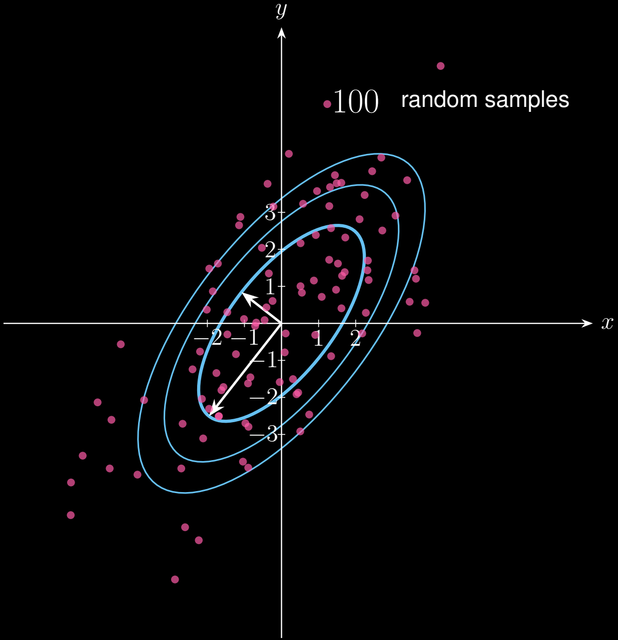


Generate random samples with $\boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $\Sigma = \begin{bmatrix} 5 & 4 \\ 4 & 7 \end{bmatrix}$

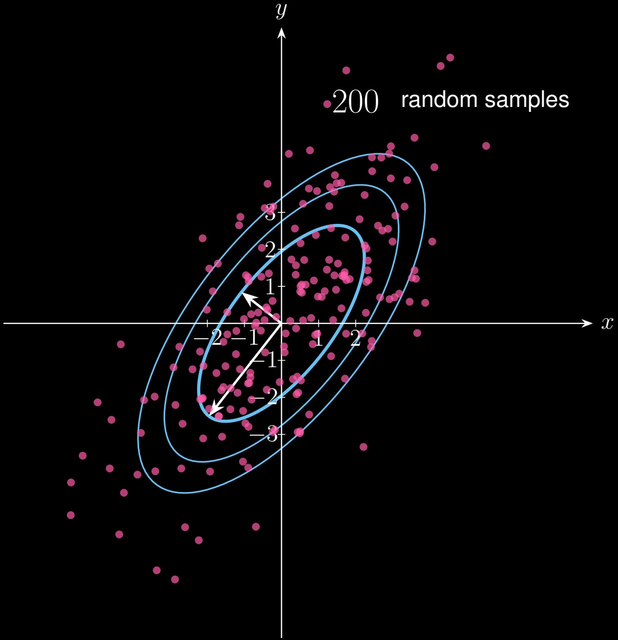
```python
import numpy as np
np.random.seed(0)

n = 1000
mu = np.zeros(2)
Sigma = np.array([[5, 4], [4, 7]])
mat = np.random.multivariate_normal(mu, Sigma, n)
np.savetxt('sample{}.txt'.format(n), mat)
```

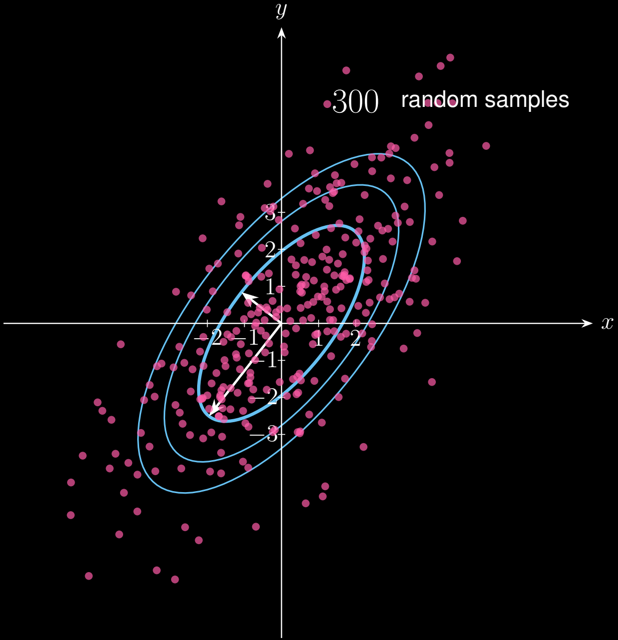


y
x
3
2
1
-2
-1
1
2
-1
-2
-3
50 random samples

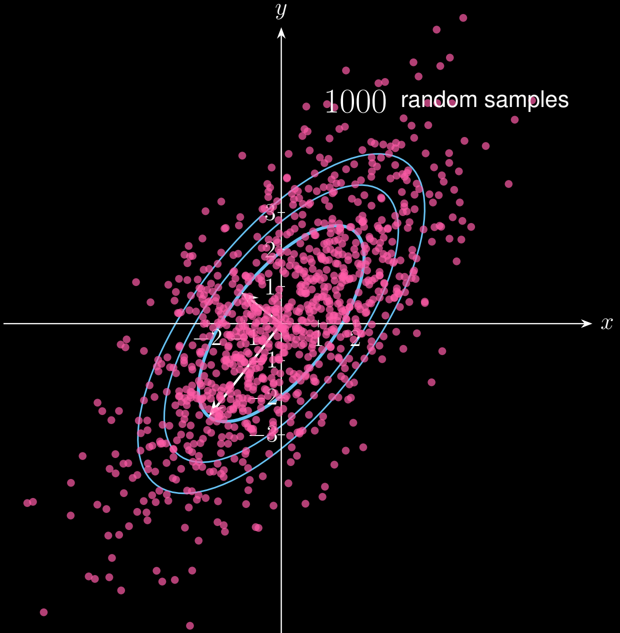


y
x
100 random samples
3
2
1
−2
−1
1
2
−1
−2
−3


y
x
200 random samples
3
2
1
-2
-1
1
2
-1
-2
-3



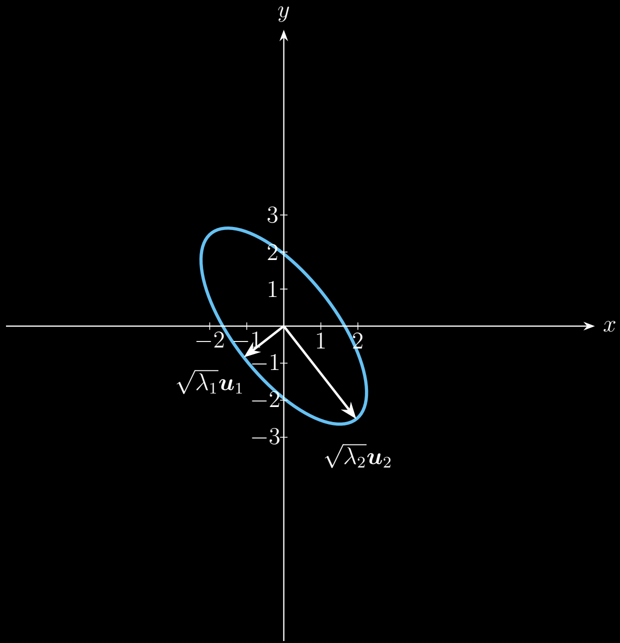
y
x
300 random samples



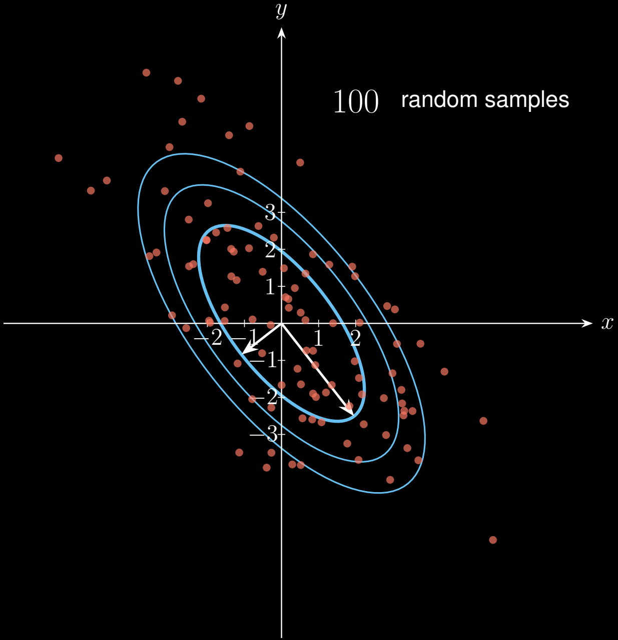
1000 random samples
y
x

Bivariate Gaussian distribution:

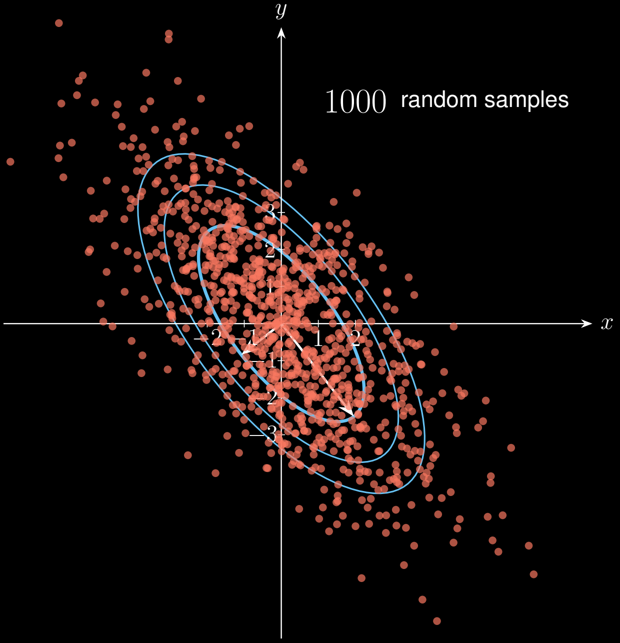
Mean vector $\boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and covariance matrix $\boldsymbol{\Sigma} = \begin{bmatrix} 5 & -4 \\ -4 & 7 \end{bmatrix}$



y
x
3
2
1
-1
-2
-3
-2
-1
1
2
$\sqrt{\lambda_1}\boldsymbol{u}_1$
$\sqrt{\lambda_2}\boldsymbol{u}_2$

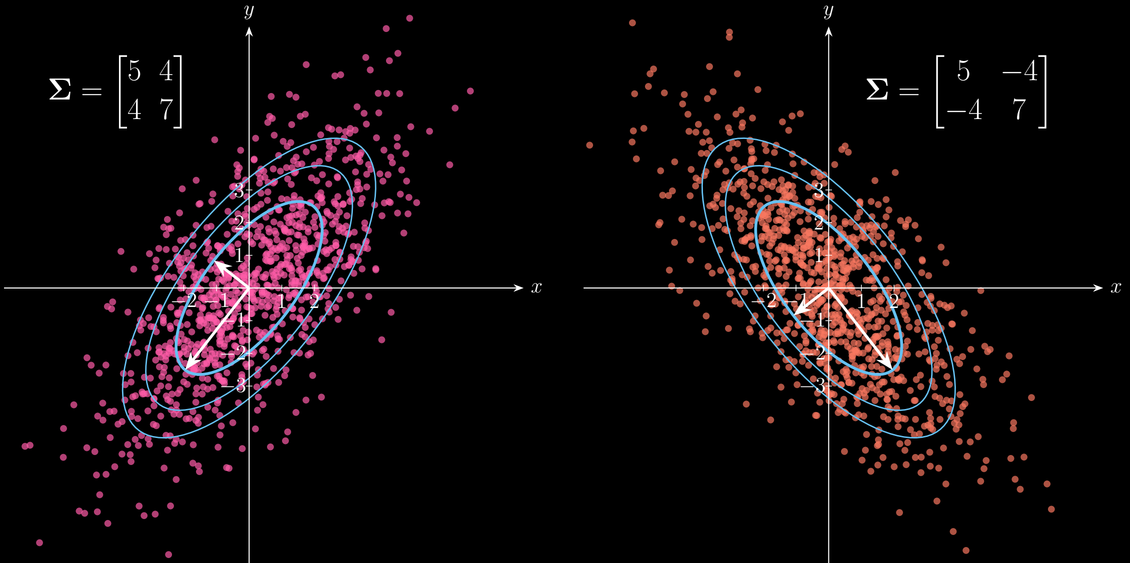


y
x
100 random samples
3
2
1
-2
-1
1
2
-1
-2
-3



1000 random samples
y
x

Eigenvalue Decomposition Defines the Shape of Multivariate Gaussian



Thanks for your attention!

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